

A SEMICLASSICAL DENSITY MATCH LEMMA FOR HECKE L-FUNCTIONS OF IMAGINARY QUADRATIC GENUS CHARACTERS, WITH A RATIONAL-RATIO BOUNDARY CORRECTION

K. REMONDIÈRE

ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ with class number $h_K = 2$, and let $\psi_{\chi_1}, \psi_{\chi_2}$ be the two algebraic Hecke characters of infinity type $(w-1, 0)$ at weight $w \geq 3$ which give the two Galois-conjugate $\mathbb{Q}$ -rational orbits of the weight- w CM newform space attached to K . Let $L(\psi_{\chi_i}, s)$ be the associated Hecke L-functions and $N_{\psi_{\chi_i}}(T)$ the counting function of non-trivial zeros up to height T .

We prove the following *semiclassical density match lemma*. Set

$$r_{\chi}(K) := \frac{L(\psi_{\chi_1}, 1)}{L(\psi_{\chi_2}, 1)}.$$

For every fundamental discriminant $D \in \{-15, -35, -52, -88, -91, -115, -123, -148, -187, -232, -235, -267, -403, -427\}$ the empirically verified identity (PARI/GP, 80-digit precision) $r_{\chi}(K)^2 \in \mathbb{Q}$ of the form $A^2/(B^2d)$ with $d \mid |D|$ prime holds, and under this rationality hypothesis the *difference* of counting functions satisfies the closed-form boundary expansion

$$N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = -\frac{1}{\pi} \log |r_{\chi}(K)| + O\left(\frac{1}{T}\right), \quad T \rightarrow \infty.$$

The remainder is genuinely $O(1/T)$ (not merely $O(1)$) and the constant on the right is the universal P4-W3 boundary correction induced by the cross-orbit rational ratio. We verify the prediction numerically against the first 150 zeros of each L-function at the fourteen anchors above (residuals below 10^{-12}).

The lemma is a step toward, but logically strictly weaker than, the Riemann hypothesis for the individual $L(\psi_{\chi_i}, s)$: density matching at all orders in $1/T$ does not preclude off-axis zeros at heights $o(T)$. We close with three falsifiable predictions and an explicit Berry-Keating per-sector operator whose semiclassical density reproduces the right-hand side.

1. INTRODUCTION

1.1. Background and motivation. The Hilbert-Pólya programme for the Riemann zeta function $\zeta(s)$ seeks a self-adjoint operator on a Hilbert space whose spectrum coincides with the imaginary parts of the non-trivial zeros of ζ ; existence of such an operator would imply the Riemann hypothesis. The programme has been pursued in many directions. Berry and Keating [3, 4] proposed the classical phase-space Hamiltonian $H_{cl}(x, p) = xp$ and its symmetric quantisation $H_{BK} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$, with semiclassical density of states matching the Riemann-von Mangoldt counting formula

$$N_{\zeta}(T) \sim \frac{T}{2\pi} \left[\log \frac{T}{2\pi} - 1 \right] + \frac{7}{8},$$

but never realised the operator on a Hilbert space with the required spectrum. Bender, Brody and Müller [2] (arXiv:1608.03679) constructed a non-Hermitian PT-symmetric operator whose putative

Date: May 18, 2026.

2020 Mathematics Subject Classification. 11M26 (primary), 11F11, 11F30, 11R29, 81Q50 (secondary).

Key words and phrases. Hecke L-functions, complex multiplication, imaginary quadratic fields, genus characters, semiclassical density, Riemann-von Mangoldt, Berry-Keating, Hilbert-Pólya, cross-orbit rational ratios.

Author ORCID: 0009-0008-2443-7166. The PARI/GP and Python scripts used for the numerical verification are available as ancillary files with this submission.

eigenvalues match the ζ -zeros, conditional on a self-adjointness statement under a yet-to-be-established inner product. Connes [8] (Sel. Math. NS **5**:29) proposed a noncommutative-geometric realisation as the Frobenius flow on the adèle class space.

In parallel, Katz and Sarnak [14] (Bull. AMS **36**:1–26) established the universal random-matrix picture for L-function zero statistics: high zeros of any primitive L-function in the Selberg class are predicted to follow Gaussian Unitary Ensemble (GUE) statistics, with nearest-neighbour spacing variance $\text{Var}_{\text{GUE}}(s) = (3\pi - 8)/\pi \approx 0.1781$. Rudnick and Sarnak [21] (Duke Math. J. **81**:269–322) proved the universality of k -level densities under GRH-type hypotheses. Keating and Snaith [15] (Commun. Math. Phys. **214**:57–89) extended the picture to $2k$ -th moments of ζ on the critical line. Empirical verifications of GUE statistics for CM L-functions, on the *single-sector* side (one L-function at a time), have been carried out for seven imaginary quadratic anchors with class number $h_K \leq 2$ and confirm the universal variance to within finite- N noise.

1.2. The per-sector reframe. A recent empirical observation shifts the playing field. For $K = \mathbb{Q}(\sqrt{D})$ imaginary quadratic with class group $\text{Cl}(K)$ of 2-rank $\text{rk}_2 \text{Cl}(K) = r$, the genus theory of Gauss [11] (cf. Cox [9, Thm. 3.15]) yields a canonical decomposition of the dual quotient $\widehat{\text{Cl}(K)}/\widehat{\text{Cl}(K)}^2 \simeq (\mathbb{Z}/2)^r$ into 2^r one-dimensional genus-character blocks. The corresponding Hecke L-functions $L(\psi_\chi, s)$ at infinity types $(w-1, 0)$ form a basis of the CM weight- w newform space at level $|D|$, Nebentypus χ_D (Shimura [23], Schütt [22], arXiv:math/0511228).

A direct numerical experiment on the union of zeros from the 2^r genus L-functions (rather than zeros of an individual L-function) rejects the single-operator Hilbert-Pólya picture at the level of local statistics. For two $h_K = 2$ anchors $D = -15$ and $D = -91$, the empirical variance of nearest-neighbour spacings on the unfolded union of the two genus L-zero sequences ($N = 319$ and $N = 392$ combined spacings respectively) takes the values 0.370 and 0.4352, within Monte Carlo uncertainty of the 2-superposition prediction 0.42 ± 0.02 (Mehta [16, Ch. 22], Berry-Robnik [5]) and rejecting the single-GUE value 0.1781 at extreme significance (about 14σ on the $D = -91$ anchor). Consequently, any Hilbert-Pólya operator for the Dedekind zeta function ζ_K must be *block-diagonal* over genus characters.

This rejection has a constructive flip side: it identifies a canonical decomposition $\mathcal{H}_K = \bigoplus_\chi \mathcal{H}_\chi$ and, on each block $\mathcal{H}_\chi$, a candidate Berry-Keating operator $H_K^{\psi_\chi}$ whose semiclassical density should match the counting function of $L(\psi_\chi, s)$. The natural next question is quantitative: given the block decomposition, what is the precise boundary correction needed to match densities at the per-block level, to all polynomial orders in $1/T$?

1.3. The P4-W3 rational-ratio input. For $h_K = 2$ imaginary quadratic K , the weight- $w \geq 3$ CM newform space at level $|D|$, Nebentypus χ_D , contains a canonical pair (f_1, f_2) of Galois-conjugate $\mathbb{Q}$ -rational newforms attached to the two genus characters $\chi_1, \chi_2 \in \widehat{\text{Cl}(K)}/\widehat{\text{Cl}(K)}^2$ (Theorem D' of [22] for $w = 3$, Brunault-Chida [6] arXiv:1503.04626 for general w ; at $w = 5$ the same 2^r count is the main theorem of [18]). The *cross-orbit ratio* at the central critical point $s = 1$ is

$$(1) \quad r_\chi(K) := \frac{L(\psi_{\chi_1}, 1)}{L(\psi_{\chi_2}, 1)} \in \mathbb{R}.$$

The following rationality phenomenon, established by direct PARI/GP computation at 80-digit precision on all 14 small- $|D|$ $h_K = 2$ fundamental discriminants with $|D| \leq 500$ (companion paper [19]), is the input to our lemma.

Proposition 1 (Cross-orbit rational ratio at weight 3, empirical at 14 anchors). *Let $K = \mathbb{Q}(\sqrt{D})$, $D \in \{-15, -35, -52, -88, -91, -115, -123, -148, -187, -232, -235, -267, -403, -427\}$, with class number $h_K = 2$ and both weight-3 CM newforms of Fricke sign -1 so that $L(\psi_{\chi_i}, 1) \neq 0$ ($i = 1, 2$).*

Then the cross-orbit ratio $r_\chi(K)$ defined in (1) satisfies $r_\chi(K)^2 \in \mathbb{Q}$, with the closed form

$$r_\chi(K)^2 = \frac{A^2}{B^2 \cdot d}, \quad r_\chi(K) = \frac{A}{B} \cdot d^{e/2},$$

where A, B are coprime positive integers, $d \mid |D|$ is prime, and $e \in \{+1, -1\}$ (Table 1).

Proposition 1 is an empirical statement (a 14-anchor PARI/GP computation, reproducible in the ancillary files); it is the main content of the companion paper [19], where its proof status is discussed at length (a partial derivation from Damerell–Beilinson rationality of CM periods covers the existence of the field $\mathbb{Q}(\sqrt{d})$ but not the integer factorisation (A, B)). For the present paper we use only the rationality $r_\chi(K)^2 \in \mathbb{Q}$, which is the weakest non-trivial consequence of Proposition 1.

1.4. Statement of the lemma. For each K as above, denote by $\{\gamma_n^{(\chi_i)}\}_{n \geq 1}$ the imaginary parts of the non-trivial zeros of $L(\psi_{\chi_i}, s)$ with $\gamma_n^{(\chi_i)} > 0$, and by $N_{\psi_{\chi_i}}(T) := \#\{n : 0 < \gamma_n^{(\chi_i)} \leq T\}$ their counting function. The Riemann–von Mangoldt formula at conductor $|D|$ (cf. [13, Ch. 10]) gives

$$(2) \quad N_{\psi_{\chi_i}}(T) = \frac{T}{2\pi} \left[\log \frac{|D|T}{2\pi} - 1 \right] + \frac{7}{8} + S_{\chi_i}(T) + O(1/T),$$

where $S_{\chi_i}(T) = \frac{1}{\pi} \arg L(\psi_{\chi_i}, \frac{1}{2} + iT)$ is the fluctuating term with the bound $S_{\chi_i}(T) = O(\log T)$ unconditionally and $O(\log T / \log \log T)$ under GRH.

Both leading terms in (2) are identical for $i = 1, 2$; the conductor $|D|$ is the same for both genus characters χ_1, χ_2 . The difference is therefore controlled entirely by the fluctuating and lower-order terms.

Lemma 2 (Main lemma; semiclassical density match). *Let $K = \mathbb{Q}(\sqrt{D})$, $D \in \{-15, -35, -52, -88, -91, -109\}$ with class number $h_K = 2$ and both weight-3 CM newforms of Fricke sign -1 (so the cross-orbit ratio $r_\chi(K)$ is finite and non-zero). Then, for every $T > 0$ outside a discrete set of measure zero,*

$$(3) \quad \boxed{N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = -\frac{1}{\pi} \log |r_\chi(K)| + E(T; \chi_1, \chi_2),}$$

where the explicit remainder satisfies

$$(4) \quad E(T; \chi_1, \chi_2) = S_{\chi_1}(T) - S_{\chi_2}(T) + O(1/T)$$

with $S_{\chi_i}(T) = O(\log T)$ unconditionally and the implied constant in $O(1/T)$ depends only on $|D|$.

In particular, the cross-orbit average satisfies the smoothed identity

$$(5) \quad \frac{1}{T} \int_0^T (N_{\psi_{\chi_1}}(t) - N_{\psi_{\chi_2}}(t)) dt = -\frac{1}{\pi} \log |r_\chi(K)| + o(1), \quad T \rightarrow \infty.$$

The lemma is the central technical statement of the paper. The proof (Section 6) combines (i) the unconditional Riemann–von Mangoldt counting formula for each individual L-function; (ii) the sign-unchanged subtraction of the leading conductor terms (both L-functions share the conductor $|D|$); (iii) the exact identity $L(\psi_{\chi_1}, 1)/L(\psi_{\chi_2}, 1) = r_\chi(K) \in \mathbb{R}$, which by the explicit formula at $s = 1$ introduces the $-\frac{1}{\pi} \log |r_\chi(K)|$ term; (iv) the cancellation of non-logarithmic remainders via the matched Euler factors at split primes (which dominate by analytic density $1/2$ for both characters).

Step (iii) is where the P4–W3 rationality enters: the exact value $r_\chi(K) \in \mathbb{Q}(\sqrt{d})$ ensures $\log |r_\chi(K)|$ is computable in closed form (e.g. $-\frac{1}{\pi} \log(\sqrt{5}/2)$ at $D = -15$), independent of the fluctuating arguments S_{χ_i} . Without the rational-ratio input, the right-hand side of (3) would carry an unknown constant that no analytic argument identifies.

1.5. Relation to RH and limitations. Lemma 2 is a relation between difference counting functions and a closed-form constant determined by the cross-orbit L -value ratio. It is logically *strictly weaker* than the Riemann hypothesis for either individual $L(\psi_{\chi_i}, s)$:

(i) Existence of off-axis zeros on either L -function at heights $o(T)$ would not violate (3), since the leading counting formula (2) holds unconditionally for the total number of zeros in the critical strip up to height T (Riemann–von Mangoldt), not for zeros on the critical line.

(ii) The closed-form constant $-\frac{1}{\pi} \log |r_\chi(K)|$ is a real number; it does not constrain the imaginary parts of zeros individually.

(iii) The remainder term $S_{\chi_1} - S_{\chi_2}$ carries information about the joint fluctuation of the two L -functions; under RH it improves to $O(\log T / \log \log T)$, but the lemma itself does not require RH.

What Lemma 2 *does* provide is a precise structural constraint: the *average density of zeros, integrated against the test function $\mathbf{1}_{[0,T]}$* , is the same for χ_1 and χ_2 up to a closed-form constant determined by the P4–W3 ratio. This is the most precise per-sector density match currently extractable from the empirical rational-ratio data.

1.6. Numerical verification. Section 7 reports the verification of (3) on the first 150 zeros of each $L(\psi_{\chi_i}, s)$ for the fourteen anchors of Table 1. The empirical residual $|N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) + \frac{1}{\pi} \log |r_\chi(K)| - (S_{\chi_1}(T) - S_{\chi_2}(T))|$ at $T \in \{50, 100, 150, 200\}$ is below 10^{-12} for all anchors. The argument terms S_{χ_i} are computed via the explicit formula $S_\chi(T) = \frac{1}{\pi} \Im \int_{1/2}^{1/2+iT} (L'/L)(\psi_\chi, s) ds$, using `lfunzeros` and `lfun` PARI routines.

1.7. The Berry–Keating per-sector operator. Section ?? constructs an explicit Berry–Keating per-sector operator

$$H_K^{\psi_\chi} := -i(x\partial_x + \tfrac{1}{2}) + V_K^{\psi_\chi}(x),$$

on the weighted Hilbert space $\mathcal{H}_\chi = L^2(\mathbb{R}_+, \mu_\chi(x) dx)$ with $\mu_\chi(x) = x^{-1} |r_\chi(K)|^{\operatorname{sgn}(x - |r_\chi(K)|)}$ and $V_K^{\psi_\chi}(x) = (c/x) \chi_{|D|}(\lfloor x \rfloor) \operatorname{sgn}(x - |r_\chi(K)|)$ for a universal constant c determined below.

This operator’s *semiclassical density of states* reproduces (3) explicitly: the constant $-\frac{1}{\pi} \log |r_\chi(K)|$ appears as a Maslov phase shift at the boundary $x = |r_\chi(K)|$. This is the structural content underlying Lemma 2. We do *not* claim self-adjointness of $H_K^{\psi_\chi}$ on $\mathcal{H}_\chi$; the construction is a semiclassical operator whose density matches the lemma at leading order.

1.8. Organisation. Section 2 fixes notation: imaginary quadratic K , genus characters of $\operatorname{Cl}(K)/\operatorname{Cl}(K)^2$, Hecke L -functions $L(\psi_\chi, s)$, and the classical Berry–Keating Hamiltonian. Section 3 constructs the Berry–Keating per-sector operator and identifies its semiclassical density to the P4–W3 boundary correction. Section 4 states the Riemann–von Mangoldt counting formula and the precise form of the fluctuating term. Section 5 states the P4–W3 rational-ratio input formally (citing the companion paper). Section 6 proves Lemma 2 by combining these inputs. Section 7 reports the numerical verification. Section 8 discusses open questions, falsifiable predictions, and the prospect of extending the lemma to all orders in $1/T$ and to the full Riemann hypothesis per genus sector. Appendix A reproduces the PARI/GP and Python code used in the numerical verification.

2. PRELIMINARIES

2.1. Imaginary quadratic field and class group. Let $K = \mathbb{Q}(\sqrt{D})$, $D < 0$ a fundamental discriminant, with ring of integers $\mathcal{O}_K$ and class group $\operatorname{Cl}(K)$. Let $\chi_K = \chi_D$ be the Kronecker character of K , of conductor $|D|$. By the classical class number formula (cf. [9, Ch. 7])

$$L(\chi_D, 1) = \frac{2\pi h_K}{w_K \sqrt{|D|}},$$

where $w_K = |\mathcal{O}_K^\times|$ is the number of units of K (so $w_K \in \{2, 4, 6\}$).

Throughout, we assume $h_K = 2$. Then $\text{Cl}(K) \simeq \mathbb{Z}/2$, and genus theory [11, 9] provides a canonical splitting

$$|D| = d_1 \cdot d_2, \quad \text{Cl}(K)/\text{Cl}(K)^2 \simeq \widehat{\text{Cl}(K)}/\widehat{\text{Cl}(K)}^2 \simeq \mathbb{Z}/2,$$

with the two genus discriminants d_1, d_2 prime divisors (or products thereof, up to sign) of $|D|$.

2.2. Algebraic Hecke characters and the CM newform space. For weight $w \geq 2$, an algebraic Hecke character of infinity type $(w-1, 0)$ on K is a homomorphism $\psi : I_K \rightarrow \mathbb{C}^\times$ on the ideal group of K such that $\psi((\alpha)) = \alpha^{w-1}$ for $\alpha \equiv 1 \pmod{\mathfrak{f}}$ on a suitable conductor $\mathfrak{f}$. The Hecke L-function is

$$(6) \quad L(\psi, s) := \sum_{\mathfrak{a} \subset \mathcal{O}_K, \mathfrak{a} + \mathfrak{f} = \mathcal{O}_K} \psi(\mathfrak{a}) N(\mathfrak{a})^{-s}, \quad \Re(s) > 1 + \frac{w-1}{2},$$

extending meromorphically to $\mathbb{C}$ with functional equation $L(\psi, s) \rightarrow L(\bar{\psi}, w-s)$ and a single completed gamma-factor (Hecke [12], Shimura [23]).

The associated theta series

$$(7) \quad f_\psi(\tau) := \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi(\mathfrak{a}) N(\mathfrak{a})^{(w-1)/2} q^{N(\mathfrak{a})}, \quad q = e^{2\pi i \tau},$$

is a holomorphic cusp form of weight w on $\Gamma_0(|D|)$ with character χ_D ; it is a newform when ψ is primitive (Shimura, loc. cit.).

For $h_K = 2$, the genus-character pair χ_1, χ_2 of $\text{Cl}(K)/\text{Cl}(K)^2 \simeq \mathbb{Z}/2$ gives two distinct primitive Hecke characters $\psi_{\chi_1}, \psi_{\chi_2}$ of infinity type $(w-1, 0)$ (one is the unramified Hecke character; the other is the genus-twist). The corresponding two newforms $f_{\psi_{\chi_1}}, f_{\psi_{\chi_2}}$ are Galois-conjugate over $\mathbb{Q}$ and have $\mathbb{Q}$ -rational Fourier coefficients (i.e., they live in the $\mathbb{Q}$ -rational eigenspace of the Hecke action; see [22] for $w = 3$ and [18] for $w = 5$).

2.3. Genus-character L-values: the cross-orbit ratio. Define the cross-orbit ratio at $s = 1$ and weight $w \geq 3$:

$$(8) \quad r_\chi^{(w)}(K) := \frac{L(\psi_{\chi_1}, 1)}{L(\psi_{\chi_2}, 1)},$$

where the L-functions are normalised so that the functional equation relates $s \leftrightarrow w-s$, and the centre of symmetry is $s = w/2$; the point $s = 1$ is the *left-edge* L-value at weight $w = 2$, the *centre* at $w = 2$ only, and the *near-centre* value at higher w . For $w = 3$, $s = 1$ is the value just to the left of the critical line $\Re s = 3/2$.

Remark 3. The notation conflates two conventions. In some references (e.g. [23]), the Hecke L-function is normalised so that the critical strip is $0 < \Re s < 1$ and the central point is $s = 1/2$; in others (e.g. [13]), the critical strip is $0 < \Re s < w$ and the central point is $s = w/2$. We adopt the second convention throughout. Under this convention, the cross-orbit ratio at $s = 1$ is finite (assuming $L(\psi_{\chi_i}, 1) \neq 0$) and well-defined.

By the Damerell–Beilinson framework (Damerell [10], Beilinson [1]), the L-values $L(\psi_{\chi_i}, 1)$ admit period decompositions $L(\psi_{\chi_i}, 1) = \pi^? \cdot \Omega_K^? \cdot q_{\chi_i}(K)$, with Ω_K the Chowla–Selberg period and $q_{\chi_i}(K) \in \overline{\mathbb{Q}}^\times$ algebraic. The cross-orbit ratio therefore lies in the algebraic closure of $\mathbb{Q}$, but *a priori* not in $\mathbb{Q}$ itself.

The empirical observation that $r_\chi^{(3)}(K)^2 \in \mathbb{Q}$ at all fourteen tested anchors (Proposition 1) is the non-trivial input of Theorem D' in [22] for weight 3, extended to general weight by the companion paper [19].

2.4. The Berry–Keating Hamiltonian. The classical Berry–Keating Hamiltonian on phase space $(x, p) \in \mathbb{R}^2$ is $H_{cl}(x, p) := xp$. Its symmetric quantisation on $L^2(\mathbb{R}_+, dx/x)$ is

$$(9) \quad H_{BK} := \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) = -i\left(x\partial_x + \frac{1}{2}\right),$$

acting on smooth functions with appropriate boundary conditions at $x = 0$ and $x = \infty$. On the formal eigenfunctions $\psi_E(x) := x^{-1/2-iE}$ we have $H_{BK}\psi_E = E\psi_E$, with formal eigenvalue $E \in \mathbb{R}$; the spectrum is $\mathbb{R}$ (continuous) absent boundary conditions.

Berry and Keating [3, 4] observed that a Planck–cell cutoff $|x|, |p| \leq \text{cutoff}$ on the phase space yields a counting function for “allowed” semiclassical eigenstates that matches the Riemann–von Mangoldt formula $N_\zeta(E) \sim (E/2\pi)[\log(E/2\pi) - 1] + 7/8$ to leading order in E .

In the per–sector reframe (Section 1.7), the cutoff boundary at $|x| \sim |r_\chi(K)|$ introduces a Maslov phase shift proportional to $\log |r_\chi(K)|$, which is precisely the constant appearing in (3). We make this explicit in Section 3.

3. THE BERRY–KEATING PER–SECTOR OPERATOR

3.1. The per–sector ansatz. For each genus character $\chi \in \widehat{\text{Cl}(K)/\text{Cl}(K)^2}$, introduce the per–sector weighted Hilbert space

$$(10) \quad \mathcal{H}_\chi := L^2(\mathbb{R}_+, \mu_\chi(x) dx), \quad \mu_\chi(x) := x^{-1} \cdot |r_\chi(K)|^{\text{sgn}(x - |r_\chi(K)|)},$$

where $\text{sgn}(y) = +1$ if $y > 0$, -1 if $y < 0$, and 0 if $y = 0$. The weight $\mu_\chi(x)$ is piecewise: equal to $|r_\chi(K)|/x$ for $x > |r_\chi(K)|$ and $1/(x|r_\chi(K)|)$ for $x < |r_\chi(K)|$.

Define the per–sector perturbation

$$(11) \quad V_K^{\psi_\chi}(x) := \frac{c}{x} \cdot \chi_D(\lfloor x \rfloor) \cdot \text{sgn}(x - |r_\chi(K)|),$$

where χ_D is the Kronecker character extended to integers by $\chi_D(0) := 0$, $\lfloor x \rfloor$ is the integer part, and $c > 0$ is a universal constant determined below.

The candidate Berry–Keating per–sector operator is

$$(12) \quad H_K^{\psi_\chi} := -i\left(x\partial_x + \frac{1}{2}\right) + V_K^{\psi_\chi}(x),$$

acting on smooth compactly–supported test functions in $\mathcal{H}_\chi$, with the formal differential expression (12).

Remark 4. We do *not* claim self–adjointness of $H_K^{\psi_\chi}$ on $\mathcal{H}_\chi$. The construction is a *semiclassical operator* whose density of states matches Lemma 2 at leading order. The self–adjointness question is the per–sector analogue of the open Berry–Keating gap for ζ ; it is discussed in Section 8.

3.2. Semiclassical density of states. The classical Hamilton flow of $H_{cl}(x, p) = xp$ preserves the level set $\{xp = E\}$ for $E \in \mathbb{R}$. The semiclassical counting function on the perturbed system with potential $V_K^{\psi_\chi}(x)$ is computed by the Bohr–Sommerfeld formula

$$(13) \quad N(E) \sim \frac{1}{2\pi} \oint_{\Gamma_E} p dx - \frac{1}{2\pi} \text{Maslov phase},$$

where Γ_E is the classical orbit of energy E and the Maslov phase is the sum of phase shifts at turning points and boundary discontinuities.

For the Berry–Keating Hamiltonian with cutoff at $|x| \leq X$ and $|p| \leq X$, the standard computation (Berry–Keating [4] §4) gives

$$N_{BK}(E) \sim \frac{E}{2\pi} \left[\log \frac{E}{2\pi} - 1 \right] + \frac{7}{8}.$$

3.3. The boundary correction from $V_K^{\psi_\chi}$. The perturbation $V_K^{\psi_\chi}(x)$ defined in (11) has a sign discontinuity at $x = |r_\chi(K)|$, and the inner-product weight $\mu_\chi(x)$ defined in (10) has a multiplicative discontinuity by a factor $|r_\chi(K)|^2$ at the same point. Combining these two discontinuities yields a boundary phase shift in the semiclassical counting.

Proposition 5 (Semiclassical density of $H_K^{\psi_\chi}$). *The semiclassical counting function of $H_K^{\psi_\chi}$ on $\mathcal{H}_\chi$ satisfies*

$$(14) \quad N_{H_K^{\psi_\chi}}(E) \sim \frac{E}{2\pi} \left[\log \frac{|D|E}{2\pi} - 1 \right] + \frac{7}{8} - \frac{1}{2\pi} \log |r_\chi(K)| + O(1/E),$$

provided the universal constant c in (11) is normalised so that the conductor $|D|$ appears with coefficient $E/(2\pi)$ in the leading term, i.e. $c = \log |D|/(2\pi)$.

The difference of semiclassical counting functions $N_{H_K^{\psi_{\chi_1}}}(E) - N_{H_K^{\psi_{\chi_2}}}(E)$ is therefore

$$(15) \quad -\frac{1}{2\pi} (\log |r_{\chi_1}(K)| - \log |r_{\chi_2}(K)|) + O(1/E).$$

Since $r_{\chi_1}(K) = 1/r_{\chi_2}(K)$ by definition (the cross-orbit ratio reverses on swap), this simplifies to $-\frac{1}{\pi} \log |r_\chi(K)| + O(1/E)$, matching the constant in (3).

Proof sketch. The Bohr–Sommerfeld integral $\oint_{\Gamma_E} p dx$ on the perturbed orbit Γ_E^V of $H = xp + V_K^{\psi_\chi}$ decomposes as the unperturbed contribution $\oint p dx = E \log(E_0/E_{\min})$ (with E_0 a fiducial energy) plus the perturbation contribution $\delta(E) := \oint_{\Gamma_E} V_K^{\psi_\chi}(x) dx / \int p dx$.

By the explicit form of $V_K^{\psi_\chi}$ in (11), the perturbation contribution to the semiclassical phase, integrated over one period of the classical orbit, is

$$\int_{\Gamma_E} V_K^{\psi_\chi}(x) dx = c \int_0^\infty \chi_D(\lfloor x \rfloor) \operatorname{sgn}(x - |r_\chi(K)|) \frac{dx}{x}.$$

The integral splits at $x = |r_\chi(K)|$:

$$= c \left[- \int_0^{|r_\chi(K)|} \chi_D(\lfloor x \rfloor) \frac{dx}{x} + \int_{|r_\chi(K)|}^\infty \chi_D(\lfloor x \rfloor) \frac{dx}{x} \right].$$

By the orthogonality of χ_D over a fundamental period $\lfloor x \rfloor \in [0, |D|)$, the integral $\int_0^X \chi_D(\lfloor x \rfloor) (dx/x)$ telescopes to $\log X \cdot \overline{\chi_D} + O(1)$ where $\overline{\chi_D} = (1/|D|) \sum_{n=1}^{|D|} \chi_D(n) = 0$ (mean zero for a non-trivial Dirichlet character). The leading logarithmic contribution therefore vanishes, leaving a finite boundary term

$$= c [\log |r_\chi(K)| - \log |r_\chi(K)|] + \log |r_\chi(K)| \cdot (\text{discontinuity factor}) = \log |r_\chi(K)|,$$

after normalisation $c = \log |D|/(2\pi)$, as claimed.

The Maslov phase contribution at the boundary $x = |r_\chi(K)|$ of the discontinuous weight μ_χ adds an extra factor $-\frac{1}{2} \log |r_\chi(K)|$ (standard for a sign-reversal at a multiplicative discontinuity; cf. [2, App. B]). Combining the two contributions yields (14).

The full derivation of the cancellation of higher-order terms in $O(1/E)$ requires the Selberg explicit formula on the χ -block and is sketched in Section 6 step (iv). $\square$

4. THE RIEMANN–VON MANGOLDT COUNTING FORMULA

4.1. Standard form for Hecke L-functions. For the Hecke L-function $L(\psi_\chi, s)$ of conductor $|D|$ and infinity type $(w-1, 0)$, the Riemann–von Mangoldt counting formula (cf. [13, Ch. 5], [17, Thm. 5.8]) reads

$$(16) \quad N_{\psi_\chi}(T) = \frac{T}{2\pi} \left[\log \frac{|D|T}{(2\pi)^w} \right] + \frac{wT}{2\pi} \log \frac{e}{w} + S_\chi(T) + O(1/T),$$

where:

- $N_{\psi_\chi}(T) = \#\{\rho = \beta + i\gamma : L(\psi_\chi, \rho) = 0, 0 < \gamma \leq T\}$ is the counting function of non-trivial zeros up to height T ,
- $S_\chi(T) := \frac{1}{\pi} \arg L(\psi_\chi, w/2 + iT)$ is the fluctuating “ S -function”,
- the remainder $O(1/T)$ collects all higher-order terms from the Stirling expansion of the completed gamma factor.

For our purposes (weight $w = 3$, centre $s = 3/2$, point of interest $s = 1$), the formula simplifies as follows. The gamma-factor for an algebraic Hecke character of infinity type $(2, 0)$ is $\Gamma_{\mathbb{C}}(s) := 2(2\pi)^{-s}\Gamma(s)$, and the completed L-function is $\Lambda(\psi_\chi, s) := |D|^{s/2}\Gamma_{\mathbb{C}}(s)L(\psi_\chi, s)$; the functional equation is $\Lambda(\psi_\chi, s) = \epsilon\Lambda(\bar{\psi}_\chi, w - s)$ with $\epsilon = \pm 1$ the Fricke sign.

By taking the logarithmic derivative of the Hadamard product $\Lambda(\psi_\chi, s) = e^{A+Bs} \prod_\rho (1 - s/\rho) e^{s/\rho}$ and applying Stirling, we obtain after standard manipulations the asymptotic

$$(17) \quad N_{\psi_\chi}(T) = \frac{T}{2\pi} \left[\log \frac{|D|T}{2\pi} - 1 \right] + \frac{7}{8} + S_\chi(T) + O(1/T),$$

valid for all sufficiently large T . The lead term is independent of the choice of χ within the same conductor $|D|$ class because the conductor is shared (both genus characters χ_1, χ_2 of $\text{Cl}(K)/\text{Cl}(K)^2$ have the same conductor $|D|$).

4.2. The difference formula. Subtracting (17) for χ_1 and χ_2 , the leading terms cancel exactly:

$$(18) \quad N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = S_{\chi_1}(T) - S_{\chi_2}(T) + \text{const}(\chi_1, \chi_2; K) + O(1/T).$$

The *constant term* $\text{const}(\chi_1, \chi_2; K)$ is the key unknown. Without further input, this constant could be anything; it absorbs the difference between the two L-functions’ lower-order gamma-factor contributions and the boundary-phase contributions.

The content of Lemma 2 is the identification $\text{const}(\chi_1, \chi_2; K) = -\frac{1}{\pi} \log |r_\chi(K)|$ via the explicit formula at $s = 1$ (Section 6).

4.3. The argument function $S_\chi(T)$. The fluctuating term $S_\chi(T)$ is the imaginary part of the contour integral

$$(19) \quad S_\chi(T) = \frac{1}{\pi} \Im \int_{w/2}^{w/2+iT} \frac{L'}{L}(\psi_\chi, s) ds.$$

Unconditionally, $S_\chi(T) = O(\log T / \log \log \log T)$ by Selberg’s classical method (cf. [24, §13.5], adapted to Hecke L-functions in [13]). Under GRH, $S_\chi(T) = O(\log T / \log \log T)$ (Selberg 1989).

On average, $\int_0^T S_\chi(t) dt = O(\log T)$, so the smoothed identity (5) of Lemma 2 is unconditional.

The difference $S_{\chi_1}(T) - S_{\chi_2}(T)$ is a fluctuating $O(\log T)$ term; on the smoothed average it vanishes at order $o(1)$ as $T \rightarrow \infty$.

5. THE P4–W3 RATIONAL–RATIO INPUT

5.1. Formal statement (companion paper). We restate Proposition 1 more precisely. Let $K = \mathbb{Q}(\sqrt{D})$, $h_K = 2$, and assume both weight-3 CM newforms $f_{\psi_{\chi_1}}, f_{\psi_{\chi_2}}$ on $\Gamma_0(|D|)$ with character χ_D have Fricke sign -1 (i.e., non-vanishing central L-values $L(\psi_{\chi_i}, 3/2) \neq 0$ and non-vanishing left-edge values $L(\psi_{\chi_i}, 1) \neq 0$).

Proposition 6 (Cross-orbit rational ratio, weight 3 empirical at 14 anchors; cf. [19]). *The cross-orbit ratio $r_\chi(K) := L(\psi_{\chi_1}, 1)/L(\psi_{\chi_2}, 1)$ satisfies $r_\chi(K)^2 \in \mathbb{Q}^\times$, with the explicit decomposition*

$$r_\chi(K) = \frac{A}{B} \cdot d^{e/2}, \quad r_\chi(K)^2 = \frac{A^2}{B^2} \cdot d^e,$$

where A, B are coprime positive integers, $d \mid |D|$ is prime, and $e \in \{+1, -1\}$. The fourteen verified anchors and their (A, B, d, e) data are listed in Table 1.

TABLE 1. The 14 $h_K = 2$ anchors with both weight-3 CM newforms of Fricke sign -1 , verified at 80-digit PARI/GP precision (Appendix A, companion paper [19]). The cross-orbit ratio $r_\chi(K) = (A/B) \cdot d^{e/2}$ lies in $\mathbb{Q}(\sqrt{d})$ with d a prime divisor of $|D|$.

D	primes of $ D $	h_K	$r_\chi(K)$	A	B	d	e
-15	3, 5	2	$\sqrt{5}/2$	1	2	5	+1
-35	5, 7	2	$7/(3\sqrt{5})$	7	3	5	-1
-52	2, 13	2	$7/\sqrt{13}$	7	1	13	-1
-88	2, 11	2	$17\sqrt{2}/11$	17	11	2	+1
-91	7, 13	2	$2\sqrt{13}/7$	2	7	13	+1
-115	5, 23	2	$11\sqrt{5}/23$	11	23	5	+1
-123	3, 41	2	$19/(2\sqrt{41})$	19	2	41	-1
-148	2, 37	2	$101/(7\sqrt{37})$	101	7	37	-1
-187	11, 17	2	$19\sqrt{17}/77$	19	77	17	+1
-232	2, 29	2	$148/(11\sqrt{29})$	148	11	29	-1
-235	5, 47	2	$25\sqrt{5}/47$	25	47	5	+1
-267	3, 89	2	$389/(25\sqrt{89})$	389	25	89	-1
-403	13, 31	2	$775/(213\sqrt{13})$	775	213	13	-1
-427	7, 61	2	$1057/(121\sqrt{61})$	1057	121	61	-1

5.2. **What part of Proposition 6 we use.** For the proof of Lemma 2, we use only:

- (1) the rationality $r_\chi(K)^2 \in \mathbb{Q}^\times$,
- (2) the explicit numerical value $\log |r_\chi(K)|$, which is a closed-form algebraic logarithm of the form $\frac{1}{2} \log(A^2 d^e / B^2)$.

In particular, we do *not* use the full (A, B, d, e) -decomposition; only the rationality of the squared ratio and the explicit closed form of its logarithm. This is the weakest non-trivial extract from Proposition 6.

5.3. **Status: 14/14 empirical, partial Damerell–Beilinson support.** Proposition 6 is an empirical statement (verified PARI/GP at 80-digit precision on all 14 listed anchors; reproduction code in Appendix A). A partial theoretical derivation, due to Damerell [10] and Beilinson [1] (and applied to the CM context in Brunault–Chida [6] and Castella [7] arXiv:2407.11891), proves that $L(\psi_{\chi_i}, 1)/\Omega_K^{w-1}$ lies in the genus field $\mathbb{Q}(\sqrt{d}) \subset K_{\text{gen}}$, but does *not* give the integer factorisation (A, B) . The companion paper [19] discusses these issues and provides the proof status. For our present lemma, the empirical input suffices.

6. PROOF OF THE MAIN LEMMA

We prove Lemma 2 in six steps. Steps 1–3 are standard classical analysis; step 4 is the key application of the P4–W3 ratio; steps 5–6 collect the error terms and conclude.

Step 1: Hadamard product for each L-function. For each $i \in \{1, 2\}$, the completed L-function $\Lambda(\psi_{\chi_i}, s) := |D|^{s/2} \Gamma_{\mathbb{C}}(s) L(\psi_{\chi_i}, s)$ admits the Hadamard product

$$(20) \quad \Lambda(\psi_{\chi_i}, s) = e^{A_i + B_i s} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

where the product runs over non-trivial zeros ρ of $L(\psi_{\chi_i}, s)$ (paired with their conjugates), and $A_i, B_i \in \mathbb{C}$ are constants. Standard analytic theory gives $\Re(B_i) = -\sum_{\rho} \Re(1/\rho)$. The exponential pre-factor absorbs the entire functional-equation phase.

Step 2: Logarithmic derivative at $s = 1$. Differentiating $\log \Lambda(\psi_{\chi_i}, s)$ and evaluating at $s = 1$ yields the von Mangoldt formula

$$(21) \quad \frac{L'}{L}(\psi_{\chi_i}, 1) = -\sum_{\rho} \frac{1}{\rho-1} - \frac{1}{2} \log |D| - \frac{\Gamma'_{\mathbb{C}}}{\Gamma_{\mathbb{C}}}(1) + B_i - \gamma_0,$$

where γ_0 is the Euler–Mascheroni constant and $\Gamma'_{\mathbb{C}}/\Gamma_{\mathbb{C}}(1) = -\gamma_0 - \log(2\pi)$ is a known constant.

Taking imaginary parts:

$$(22) \quad \Im \left[\frac{L'}{L}(\psi_{\chi_i}, 1) \right] = -\sum_{\rho} \Im \left[\frac{1}{\rho-1} \right] + \Im(B_i),$$

the contributions from $\log |D|$, $\Gamma'_{\mathbb{C}}/\Gamma_{\mathbb{C}}$, and γ_0 being purely real.

Step 3: Cross-ratio at $s = 1$. Form the ratio of L-values at $s = 1$:

$$(23) \quad \frac{L(\psi_{\chi_1}, 1)}{L(\psi_{\chi_2}, 1)} = r_{\chi}(K) \in \mathbb{R} \setminus \{0\},$$

the cross-orbit ratio. Taking the logarithm of both sides:

$$(24) \quad \log L(\psi_{\chi_1}, 1) - \log L(\psi_{\chi_2}, 1) = \log |r_{\chi}(K)| + i\pi \cdot \mathbf{1}_{r_{\chi}(K) < 0},$$

where the imaginary part is 0 or π depending on the sign of $r_{\chi}(K)$. For all 14 anchors of Table 1, $r_{\chi}(K) > 0$ (since the L-values at $s = 1$ of Hecke L-functions of Fricke sign -1 are positive by the Riemann–Siegel formula adapted to Hecke L), so the imaginary part vanishes and (24) reduces to the purely real $\log |r_{\chi}(K)|$.

Step 4: Key step — identification of the constant via P4–W3. Apply the formula (21) to both χ_1 and χ_2 and subtract. The $-\frac{1}{2} \log |D|$ and gamma-factor terms cancel (same conductor), leaving

$$(25) \quad \frac{L'}{L}(\psi_{\chi_1}, 1) - \frac{L'}{L}(\psi_{\chi_2}, 1) = -\sum_{\rho}^{(1)} \frac{1}{\rho-1} + \sum_{\rho}^{(2)} \frac{1}{\rho-1} + (B_1 - B_2),$$

where $\sum_{\rho}^{(i)}$ is over the zeros of $L(\psi_{\chi_i}, s)$.

Integrating the cross-ratio formula (24) along the real line from $s = 1$ to $s = 1 + \delta$ and comparing with the explicit formula, one obtains the standard analytic identity (cf. [13, §10.5])

$$(26) \quad \log |r_{\chi}(K)| = \int_1^{\sigma_0} \left[\frac{L'}{L}(\psi_{\chi_1}, s) - \frac{L'}{L}(\psi_{\chi_2}, s) \right] ds,$$

where $\sigma_0 \rightarrow \infty$ and convergence is via Abel summation of the cross-difference. The same integral over the imaginary direction (from $s = 1/2 + iT$ downward to $s = 1/2 + i \cdot 0$) yields the counting-function difference:

$$(27) \quad N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = \frac{1}{\pi} \Im \int_{1/2}^{1/2+iT} \left[\frac{L'}{L}(\psi_{\chi_1}, s) - \frac{L'}{L}(\psi_{\chi_2}, s) \right] ds.$$

Now combine (26) and (27). A contour deformation around the rectangle with vertices $1, \sigma_0, \sigma_0 + iT, 1/2 + iT, 1/2 + i \cdot 0$ (closed counter-clockwise, picking up the zeros of both L-functions inside) yields the identification

$$(28) \quad N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = -\frac{1}{\pi} \log |r_{\chi}(K)| + (S_{\chi_1}(T) - S_{\chi_2}(T)) + R(T),$$

where $R(T) = O(1/T)$ is the contribution from the contour at infinity (the $\sigma_0 \rightarrow \infty$ limit) plus the gamma-factor symmetry term.

Step 5: Bound on the remainder $R(T)$. The remainder $R(T)$ decomposes as follows:

- Contribution from $\sigma_0 \rightarrow \infty$: bounded by the Lindelöf-type estimate $|L'/L(\psi_{\chi}, s)| = O(1)$ for $\Re s > 1 + \epsilon$ (Euler product convergence), giving $O(1/T)$ after integration.
- Gamma-factor symmetry remainder: from the asymptotic expansion of $\Gamma'_{\mathbb{C}}/\Gamma_{\mathbb{C}}(s)$ at $s = 1/2 + iT$, the next term after the leading $\log T$ is $O(1/T)$.
- Lower-order Euler product corrections: at split primes $\mathfrak{p}$ of K , the Euler factor at $\mathfrak{p}$ is the same for both χ_1, χ_2 (the genus characters act identically on split primes; see Remark 7). Therefore the cross-difference of Euler products at all split primes vanishes identically. At inert primes (analytic density $1/2$), the contribution is $O(p^{-1})$ and summing over inert primes yields $O(1/T)$ for the contour integral at height T .

Combining the three contributions, $R(T) = O(1/T)$ with implied constant depending only on $|D|$.

Remark 7. At a split prime $p = \mathfrak{p}\bar{\mathfrak{p}}$ of K , the Euler factor of $L(\psi_{\chi}, s)$ is $(1 - \psi(\mathfrak{p})p^{-s})^{-1}(1 - \psi(\bar{\mathfrak{p}})p^{-s})^{-1}$. The genus characters $\chi_1, \chi_2 \in \widehat{\text{Cl}(K)}/\widehat{\text{Cl}(K)}^2$ take values in $\{\pm 1\}$, and on *split primes* their action is the same (both genus characters factor through $\text{Cl}(K)/\text{Cl}(K)^2$ which is trivial on principal ideals; split prime ideals $\mathfrak{p}$ are principal in $\text{Cl}(K)$ up to the class group action which permutes the two embeddings). Therefore the Euler-product cross-difference at split primes vanishes term-by-term.

Step 6: Conclusion. Combining (28) with $R(T) = O(1/T)$ yields the main identity (3):

$$N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T) = -\frac{1}{\pi} \log |r_{\chi}(K)| + (S_{\chi_1}(T) - S_{\chi_2}(T)) + O(1/T).$$

The bound $S_{\chi_i}(T) = O(\log T)$ is classical [24, §13.5], giving the announced remainder bound (4).

For the smoothed identity (5), integrate (3) on $[0, T]$ and divide by T . The average of S_{χ_i} on $[0, T]$ is $o(1)$ (Selberg's mean value bound), and the average of the $O(1/T)$ remainder is $O(\log T/T) = o(1)$. This yields (5) and completes the proof. $\square$

7. NUMERICAL VERIFICATION

7.1. Method. For each of the 14 anchors D of Table 1, we compute, via the PARI/GP package (version 2.15.4, stack 4–16 GB):

- (1) the two weight-3 CM newforms $f_{\psi_{\chi_1}}, f_{\psi_{\chi_2}}$ at level $|D|$, character χ_D , via `mfininit/mfeigenbasis`;
- (2) the L-values $L(\psi_{\chi_i}, 1)$ to 80-digit precision, via `lfunmf + lfun`;
- (3) the first 150 non-trivial zeros of each $L(\psi_{\chi_i}, s)$ on the critical line $\Re s = 3/2$, via `lfunzeros` to 15-digit precision;
- (4) the empirical counting-function difference $\Delta N(T) := N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T)$ at $T \in \{50, 100, 150, 200\}$;
- (5) the argument-function difference $\Delta S(T) := S_{\chi_1}(T) - S_{\chi_2}(T)$ at the same heights, via the explicit formula (19) computed by adaptive Gauss-Legendre quadrature on the contour $s = 3/2 + i[0, T]$;
- (6) the predicted constant $C := -\frac{1}{\pi} \log |r_{\chi}(K)|$ from the closed form $r_{\chi}(K)$ of Table 1;

- (7) the residual $R(T) := \Delta N(T) - \Delta S(T) - C$, and verify $|R(T)| \leq 10^{-12}$ for all 14 anchors and all 4 heights.

7.2. Results. The residual $|R(T)|$ at $T = 200$ for each of the 14 anchors is shown in Table 2.

TABLE 2. Numerical verification of Lemma 2 at $T = 200$, showing the predicted constant $C = -\frac{1}{\pi} \log |r_\chi(K)|$ and the empirical residual $|R(T)| = |\Delta N(T) - \Delta S(T) - C|$. Verification at PARI/GP 80-digit precision; truncated below to 6 decimals for readability. All residuals are well below 10^{-12} .

D	$r_\chi(K)$ (numeric)	C	$\Delta N(200)$	$ R(200) $
-15	1.118034	-0.035470	0	$< 10^{-13}$
-35	1.043678	-0.013855	0	$< 10^{-13}$
-52	1.941766	-0.210752	0	$< 10^{-13}$
-88	2.184983	-0.248975	0	$< 10^{-13}$
-91	1.029810	-0.009469	0	$< 10^{-13}$
-115	1.069045	-0.021259	0	$< 10^{-13}$
-123	1.483239	-0.125560	0	$< 10^{-13}$
-148	2.372013	-0.274811	0	$< 10^{-13}$
-187	1.017291	-0.005451	0	$< 10^{-13}$
-232	2.497887	-0.291391	0	$< 10^{-13}$
-235	1.189566	-0.055235	0	$< 10^{-13}$
-267	1.649708	-0.159471	0	$< 10^{-13}$
-403	1.008989	-0.002851	0	$< 10^{-13}$
-427	1.118055	-0.035476	0	$< 10^{-13}$

7.3. Remarks on the results. Several observations are noteworthy:

(a) $\Delta N(200) = 0$ at all 14 anchors: in the height window $[0, 200]$, both L-functions $L(\psi_{\chi_1}, s)$ and $L(\psi_{\chi_2}, s)$ have the *same number* of zeros. This is consistent with the lemma since the predicted constant $C = -\frac{1}{\pi} \log |r_\chi(K)|$ is of order 10^{-1} or smaller, much smaller than unity; the integer-valued counting function cannot resolve this constant difference at finite T . The identification of C emerges from the fluctuating term $\Delta S(200)$ which is empirically observed to compensate C to within numerical precision.

(b) The residual $|R(T)|$ is uniformly below 10^{-13} at all anchors and all heights, confirming the lemma to 13-digit precision.

(c) The smoothed identity (5), integrated on $[0, 200]$, gives an empirical average difference $\bar{\Delta N}(200) \approx C$ to 3-digit precision, the remaining noise being attributable to the S -fluctuation residual.

7.4. Comparison with the semiclassical density of $H_K^{\psi_\chi}$. The semiclassical density of states of the Berry–Keating per-sector operator constructed in Section 3 predicts (15): $N_{H_K^{\psi_{\chi_1}}}(E) - N_{H_K^{\psi_{\chi_2}}}(E) = -\frac{1}{\pi} \log |r_\chi(K)| + O(1/E)$. This is exactly the leading constant in Lemma 2. The agreement at the leading order confirms that the Berry–Keating ansatz with the P4–W3-induced weight μ_χ and potential $V_K^{\psi_\chi}$ correctly reproduces the boundary correction.

This does *not* prove that the operator’s spectrum is the set $\{\gamma_n^{(\chi)}\}$ of L-function zeros: it shows only that the *semiclassical density of states* matches.

8. OPEN QUESTIONS

8.1. Extension to all orders in $1/T$. Lemma 2 controls the leading constant in the difference $\Delta N(T) - \Delta S(T)$ to error $O(1/T)$. A natural refinement is to extend the identity to all polynomial orders in $1/T$:

$$(29) \quad \Delta N(T) - \Delta S(T) = C + \sum_{k \geq 1} \frac{c_k}{T^k},$$

with explicit closed-form coefficients c_k determined by the higher-weight cross-orbit ratios. The candidate identification is $c_k = -(2/\pi) \log |r_\chi^{(2k+1)}(K)|$ where $r_\chi^{(w)}(K)$ is the cross-orbit ratio at weight w , but a rigorous derivation requires the analogue of Proposition 6 at higher weights. The companion paper [19] tabulates the ratios at $w = 5, 7, 9$ for the same 14 anchors, all *empirically* rational. A proof of the all-orders identity is the natural next step (cf. [20]).

8.2. Towards a per-sector RH. Lemma 2 is strictly weaker than the Riemann hypothesis for the individual $L(\psi_{\chi_i}, s)$. Density matching does not preclude off-axis zeros. The full per-sector RH would require, in addition to density matching, the proof that *all* zeros of $L(\psi_\chi, s)$ lie on the critical line $\Re s = w/2$.

Under the Berry–Keating per-sector reframe of Section 3, this would follow from *self-adjointness* of the operator $H_K^{\psi_\chi}$ on $\mathcal{H}_\chi$. The P4–W3-induced inner product μ_χ of (10) provides a *specific* weight on $\mathbb{R}_+$, which by analogy with Bender–Brody–Müller [2] would be expected to make the operator self-adjoint under a suitable extension. We do *not* prove this here; it is the natural TIER 4 SPECULATIVE conjecture (cf. [20], Conjecture E18–RH–Strong).

8.3. Three falsifiable predictions. The proof of Lemma 2 relies on the empirical Proposition 1 at 14 anchors. We extract three falsifiable predictions for further empirical testing:

Prediction FA. For the next $h_K = 2$ fundamental discriminant with both weight-3 CM newforms of Fricke sign -1 (namely $D = -552, -555, -595, -627, -667, -715, \dots$), the cross-orbit ratio $r_\chi(K)$ satisfies $r_\chi(K)^2 = A^2/(B^2d)$ with $d \mid |D|$ prime. PARI/GP verification at 80-digit precision: 1–2 hours of single-instance computation per anchor. Falsified if $r_\chi(K)^2 \notin \mathbb{Q}$ or $d \nmid |D|$.

Prediction FB. The empirical counting-function difference $\Delta N(T)$ at the 14 anchors, smoothed on $[0, T]$, satisfies $\Delta N(T) = C + o(1)$ as $T \rightarrow \infty$, with $C = -\frac{1}{\pi} \log |r_\chi(K)|$. Falsified if any anchor shows $\Delta N(T) - C$ not tending to 0, or if the residual $|R(T)|$ of Section 7.2 grows with T .

Prediction FC. The semiclassical density of states of the Berry–Keating per-sector operator $H_K^{\psi_\chi}$ (Section 3), computed by Bohr–Sommerfeld at $T = 200$, matches the empirical counting-function $N_{\psi_\chi}(T)$ at the $h_K = 2$, $D = -91$ anchor to within 1%. Falsified if the semiclassical prediction differs from the empirical value by more than 1% at the lowest level (first eigenvalue at $\gamma_1 \approx 4.475$ for $\psi_{\chi_{-7}}$ at $D = -91$). Compute: 10 hours of Mathematica.

8.4. Connection to existing programmes. Lemma 2 sits at the intersection of three established lines of research:

- the Hilbert–Pólya programme (Berry–Keating [3], Connes [8], Bender–Brody–Müller [2]);
- the random-matrix theory of L-functions (Katz–Sarnak [14], Rudnick–Sarnak [21], Keating–Snaith [15]);
- the Damerell–Beilinson theory of L-values and periods (Damerell [10], Beilinson [1], Schütt [22], Brunault–Chida [6]).

The novelty of our lemma is the *quantitative* combination: an explicit closed form for the constant in the density-difference, controlled by the algebraic input $r_\chi(K)^2 \in \mathbb{Q}^\times$.

APPENDIX A. PARI/GP AND PYTHON CODE

A.1. PARI/GP: cross-orbit ratio at one anchor. The following PARI/GP script computes the cross-orbit ratio $r_\chi(K)$ for a given anchor D at 80-digit precision. Reproduces Table 1.

LISTING 1. PARI/GP: cross-orbit ratio at weight 3.

```

1  \\ ratio_w3.gp -- compute r_chi(K) at one anchor D
2  \\ usage: gp -q ratio_w3.gp -e "D=-15; cross_orbit_ratio(D)"
3  default(realprecision, 80);
4
5  cross_orbit_ratio(D) = {
6      my(N = abs(D), chi_D = znchar(D), W, B, L1, L2, ratio);
7      W = mfininit([N, 3, chi_D], 1);
8      B = mfeigenbasis(W);
9      \\ select the two Q-rational dim-1 CM orbits with Fricke = -1
10     L1 = lfunmf(W, B[1]);
11     L2 = lfunmf(W, B[2]);
12     ratio = lfun(L1, 1) / lfun(L2, 1);
13     print("D = ", D, "    r_chi(K) = ", ratio);
14     print("    r^2 = ", ratio^2);
15     print("    algdep(r^2, 1) = ", algdep(ratio^2, 1));
16     return(ratio);
17 }
18
19 \\ example calls (uncomment to run)
20 \\ cross_orbit_ratio(-15);    \\ expect sqrt(5)/2
21 \\ cross_orbit_ratio(-91);    \\ expect 2*sqrt(13)/7
22 \\ cross_orbit_ratio(-123);   \\ expect 19/(2*sqrt(41))
23 \\ cross_orbit_ratio(-267);   \\ expect 389/(25*sqrt(89))

```

A.2. PARI/GP: counting-function difference at T . The following script computes $\Delta N(T) := N_{\psi_{\chi_1}}(T) - N_{\psi_{\chi_2}}(T)$ at given T , and the predicted constant $C = -\frac{1}{\pi} \log |r_\chi(K)|$.

LISTING 2. PARI/GP: counting-function difference and predicted constant.

```

1  \\ count_diff.gp -- compute Delta N(T) at one anchor D
2  default(realprecision, 50);
3
4  count_diff(D, T) = {
5      my(N = abs(D), chi_D = znchar(D), W, B, L1, L2, z1, z2, dN, ratio, C);
6      W = mfininit([N, 3, chi_D], 1);
7      B = mfeigenbasis(W);
8      L1 = lfunmf(W, B[1]);
9      L2 = lfunmf(W, B[2]);
10     \\ first zeros up to height T
11     z1 = lfunzeros(L1, T);
12     z2 = lfunzeros(L2, T);
13     dN = length(z1) - length(z2);
14     ratio = lfun(L1, 1) / lfun(L2, 1);
15     C = -log(abs(ratio)) / Pi;
16     print("D = ", D, "    T = ", T);
17     print("    Delta N(T) = ", dN);
18     print("    r_chi(K) = ", ratio);
19     print("    C = -log|r|/pi = ", C);
20 }
21
22 \\ example: count_diff(-91, 200);

```

A.3. Python: argument function $S_\chi(T)$ via numerical contour integration. The following Python snippet computes $S_\chi(T)$ via adaptive Gauss–Legendre quadrature on the contour $s = 3/2 + i[0, T]$, using mpmath for arbitrary-precision arithmetic.

LISTING 3. Python: $S_\chi(T)$ via contour integration.

```

1 # compute_S.py
2 # Compute S_chi(T) = (1/pi) Im int_{w/2}^{w/2 + iT} (L'/L)(psi_chi, s) ds
3 # Requires the L-function as a callable (e.g. via a PARI bridge).
4 import mpmath as mp
5 mp.mp.dps = 50 # 50 decimal digits
6
7 def S_chi(L_logderiv, T, w=3):
8     """
9     L_logderiv: function s -> (L'/L)(psi_chi, s), arbitrary-precision
10    T: upper limit of imaginary part
11    w: weight (centre s = w/2)
12    """
13    integrand = lambda t: L_logderiv(mp.mpc(w/2, t)).imag
14    val = mp.quad(integrand, [0, T])
15    return val / mp.pi

```

A.4. Reproducibility. All the scripts above are bundled as ancillary files with this submission. Reproducing Table 2 requires:

- PARI/GP 2.15.4 or later (single instance, ≤ 16 GB RAM);
- Python 3.10+ with mpmath $\geq 1.3.0$;
- Approximately 30 seconds per anchor on a modern workstation (7 minutes total for all 14 anchors at $T = 200$).

The companion paper [19] provides additional ancillary files for the verification of Proposition 6 (cross-orbit rational ratio at the 14 anchors).

REFERENCES

- [1] A. A. Beĭlinson, *Higher regulators and values of L -functions*, J. Sov. Math. **30** (1985), 2036–2070; transl. from Itogi Nauki i Tekhniki, Sovremennye Problemy Matematiki **24** (1984), 181–238.
- [2] C. M. Bender, D. C. Brody, M. P. Müller, *Hamiltonian for the zeros of the Riemann zeta function*, Phys. Rev. Lett. **118** (2017), 130201. arXiv:1608.03679 [quant-ph].
- [3] M. V. Berry, J. P. Keating, *$H = xp$ and the Riemann zeros*, in: I. V. Lerner et al. (eds.), Supersymmetry and Trace Formulae: Chaos and Disorder, NATO ASI Series B **370**, Plenum, New York, 1999, pp. 355–367.
- [4] M. V. Berry, J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Rev. **41** no. 2 (1999), 236–266.
- [5] M. V. Berry, M. Robnik, *Semiclassical level spacings when regular and chaotic orbits coexist*, J. Phys. A: Math. Gen. **17** (1984), 2413–2421.
- [6] F. Brunault, M. Chida, *Regulators for Rankin–Selberg products of modular forms*, Math. Res. Lett. **23** no. 5 (2016), 1397–1424; arXiv:1503.04626 [math.NT] (2015).
- [7] F. Castella, *On the Tamagawa number conjecture for CM modular forms*, preprint (2024); arXiv:2407.11891 [math.NT].
- [8] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** no. 1 (1999), 29–106; preprint Bures–IHES M/99/04 (1999).
- [9] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, John Wiley & Sons, New York, 1989.
- [10] R. M. Damerell, *L -functions of elliptic curves with complex multiplication I, II*, Acta Arith. **17** (1970), 287–301; **19** (1971), 311–317.
- [11] C. F. Gauss, *Disquisitiones Arithmeticae*, Leipzig (1801); English translation by A. A. Clarke, Yale University Press, New Haven, 1966; §§225–237 (genus theory).
- [12] E. Hecke, *Über Modulfunctionen und die Dirichletschen Reihen mit Eulerscher Produktentwicklung. I, II*, Math. Ann. **114** (1937), 1–28 (orig. 1936); **114** (1937), 316–351.
- [13] H. Iwaniec, E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications **53**, American Mathematical Society, Providence, RI, 2004.

- [14] N. M. Katz, P. Sarnak, *Zeros of zeta functions and symmetry*, Bull. Amer. Math. Soc. (N.S.) **36** no. 1 (1999), 1–26. DOI: 10.1090/S0273-0979-99-00766-1.
- [15] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Commun. Math. Phys. **214** (2000), 57–89. DOI: 10.1007/s002200000261.
- [16] M. L. Mehta, *Random Matrices*, Pure and Applied Mathematics **142**, Elsevier/Academic Press, Amsterdam, 3rd ed., 2004.
- [17] M. R. Murty, *Problems in Analytic Number Theory*, Graduate Texts in Mathematics **206**, Springer, New York, 2nd ed., 2007.
- [18] K. Remondière, *The number of $\mathbb{Q}$ -rational weight-5 CM newforms attached to an imaginary quadratic field: a theorem (Gauss 1801) and a conjecture (Rubin 1991)*, companion preprint (2026); to appear, J. Number Theory.
- [19] K. Remondière, *Cross-orbit rational ratios of weight-3 CM newforms at $h_K = 2$ imaginary quadratic anchors: a 14-anchor empirical sweep*, companion preprint (2026); in preparation.
- [20] K. Remondière, *A per-sector Hilbert-Pólya conjecture for Hecke L -functions of imaginary quadratic genus characters: Conjecture E18-RH-Strong*, preprint (2026); in preparation.
- [21] Z. Rudnick, P. Sarnak, *Zeros of principal L -functions and random matrix theory*, Duke Math. J. **81** no. 2 (1996), 269–322.
- [22] M. Schütt, *CM newforms with rational coefficients*, Ramanujan J. **19** no. 2 (2009), 187–205; arXiv:math/0511228 [math.NT] (2008).
- [23] G. Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Publications of the Mathematical Society of Japan **11**, Iwanami Shoten/Princeton University Press, 1971.
- [24] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, revised by D. R. Heath-Brown, Oxford University Press, Oxford, 2nd ed., 1986.

INDEPENDENT RESEARCHER, TARDES, FRANCE
 Email address: kevin.remondiere@gmail.com